

Unit 1

PROSPERITY, INEQUALITY, AND PLANETARY LIMITS

OUTLINE

A. Introduction

B. Topic 1: Data on changes over time

C. Topic 2: Explaining the hockey stick

D. Topic 3: Further important considerations

A. Introduction

Most pressing problem poll...

- What is the most pressing problem economists should address?

... word cloud

2024 Global word cloud



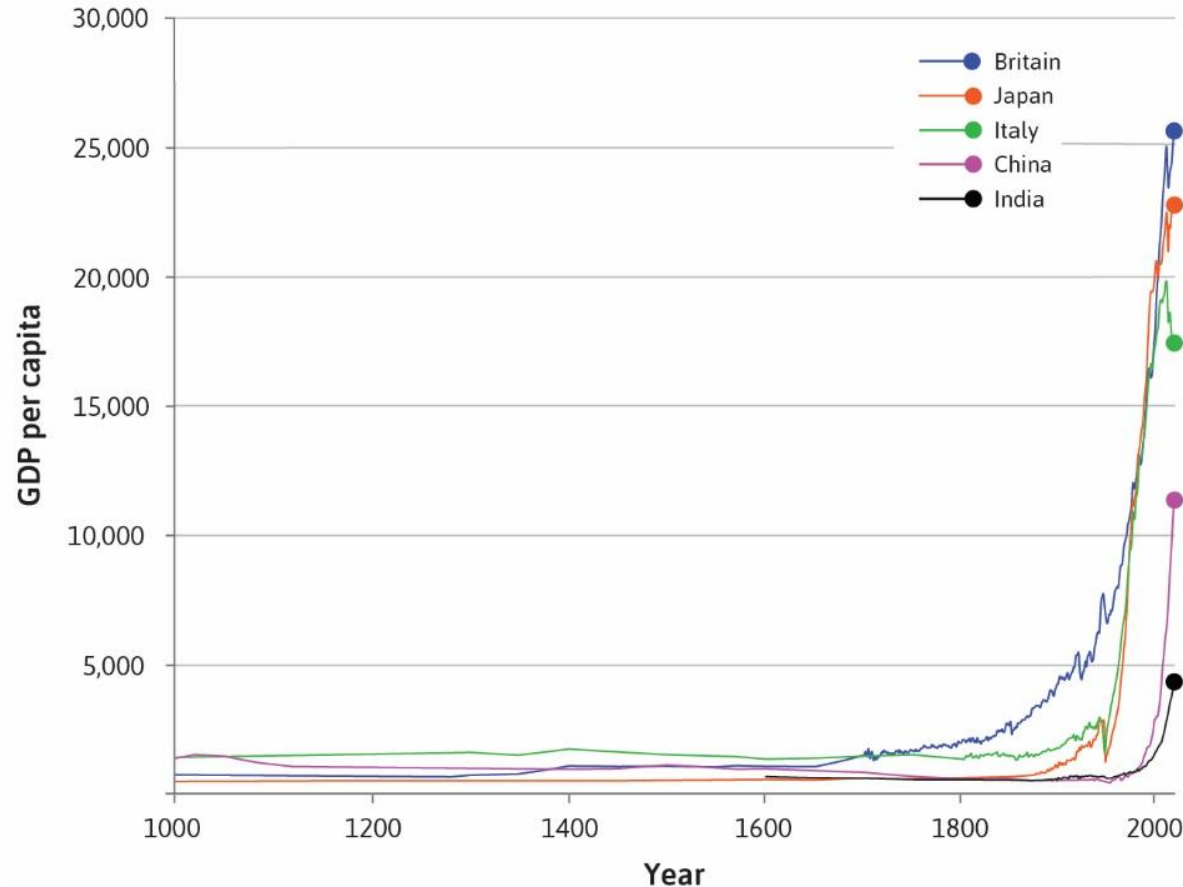
From [coreecon](https://coreecon.com)

The Context for This Unit

Data

Production

Capitalism



Q: What are the elements of a graph that we need to look at to understand the "story" that the graph is telling?

Rapid, sustained growth in average living standards since 1700.
How did this happen?

B. Topic 1: Data on changes over time

Measuring income and living standards

Gross Domestic Product (GDP) = A measure of the market value of the output of final goods and services in the economy in a given period. Output of intermediate goods that are inputs to final production is excluded to prevent double counting.

- Usually expressed in **per-capita** terms (as an average income).

GDP per capita $\neq$ Disposable income

Disposable income = Total income – taxes + government transfers

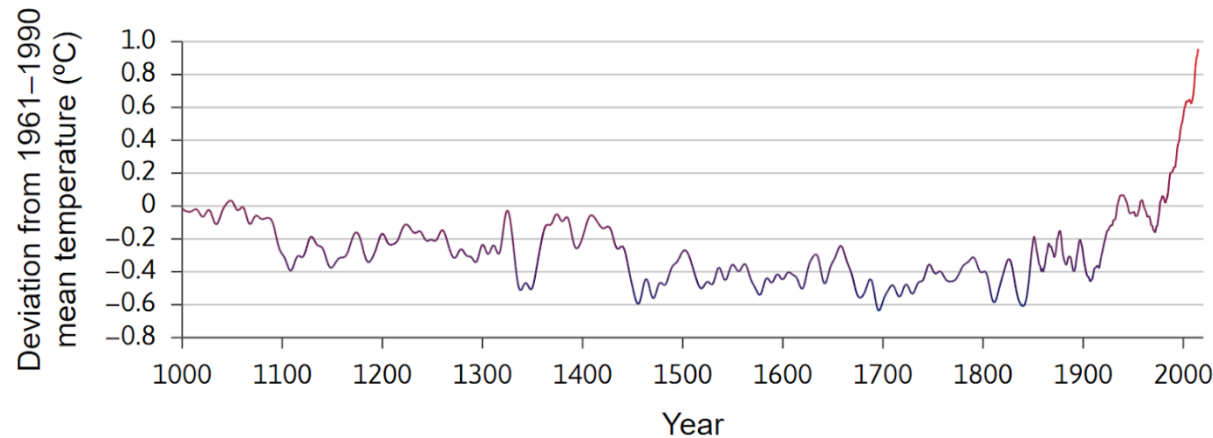
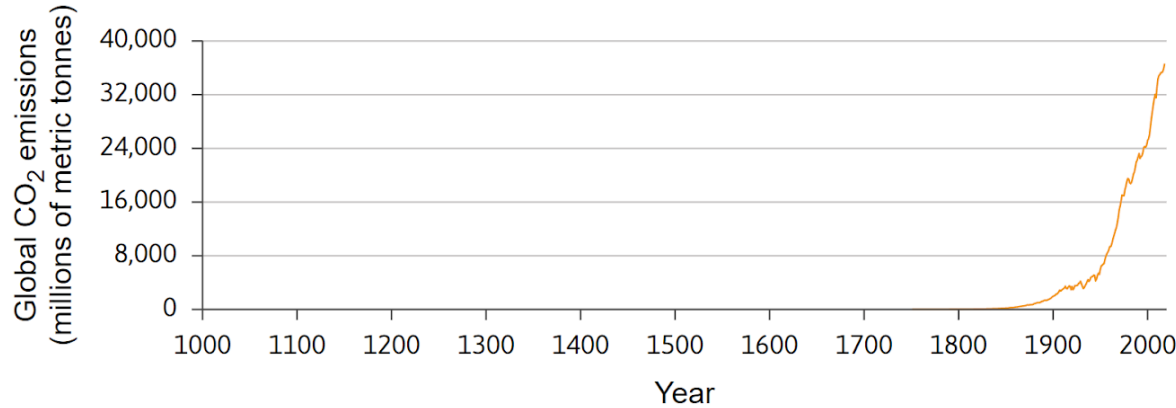
GDP: An incomplete measure

- GDP tells us how big the the total output of a country is
- GDP per capita tells us how large each person's slice of that output is if everyone received the same amount.
- Consider a group of people whose income is \$5,000 per month. Now imagine half of the group has their income increased to \$9,000 and the other half has their income reduced to \$1,000. Average income is unchanged but what about living standards?
- Does income distribution affect wellbeing?

GDP and depletion of natural resources

- The failure to count the depletion of our natural surroundings can make a big difference in our assessment of average living standards.
- Between 1971 and 1984 the annual rate of growth of Indonesian GDP was 7.1%. If the depletion of petroleum reserves, forests, and soil is taken into account, then it would have been 4%.
- In order to count output in a way that truly informs us about how well off people are we must not only include what has been used up, but also the depletion of natural resources.

Climate change



A: Form pairs and describe to each other what the graph is showing.

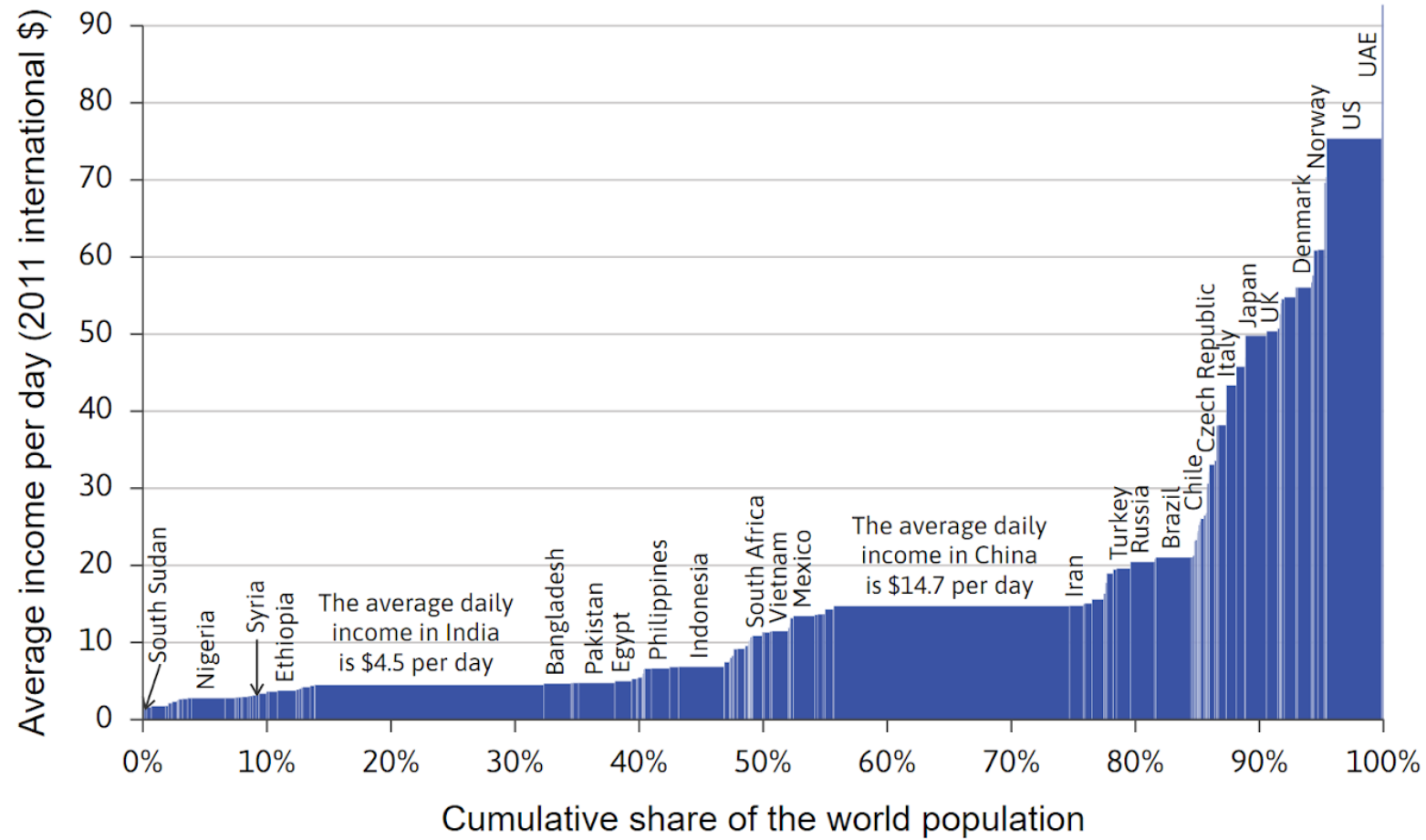
Share your interpretation with the class:

- What do you notice? What do you wonder?

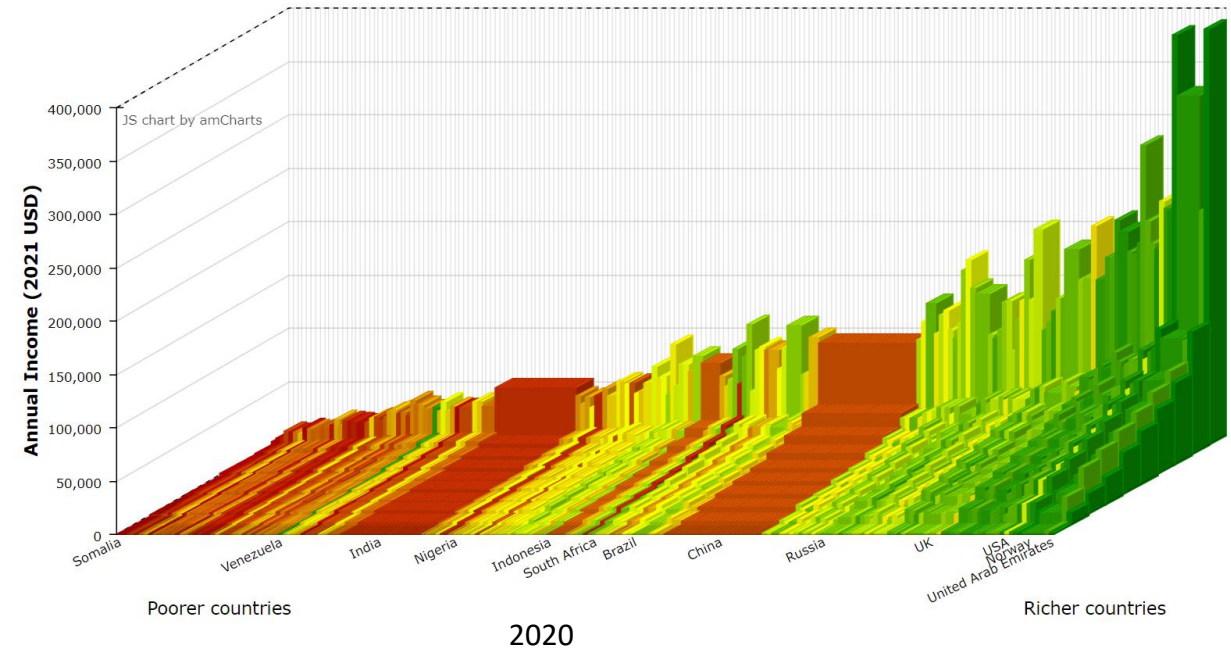
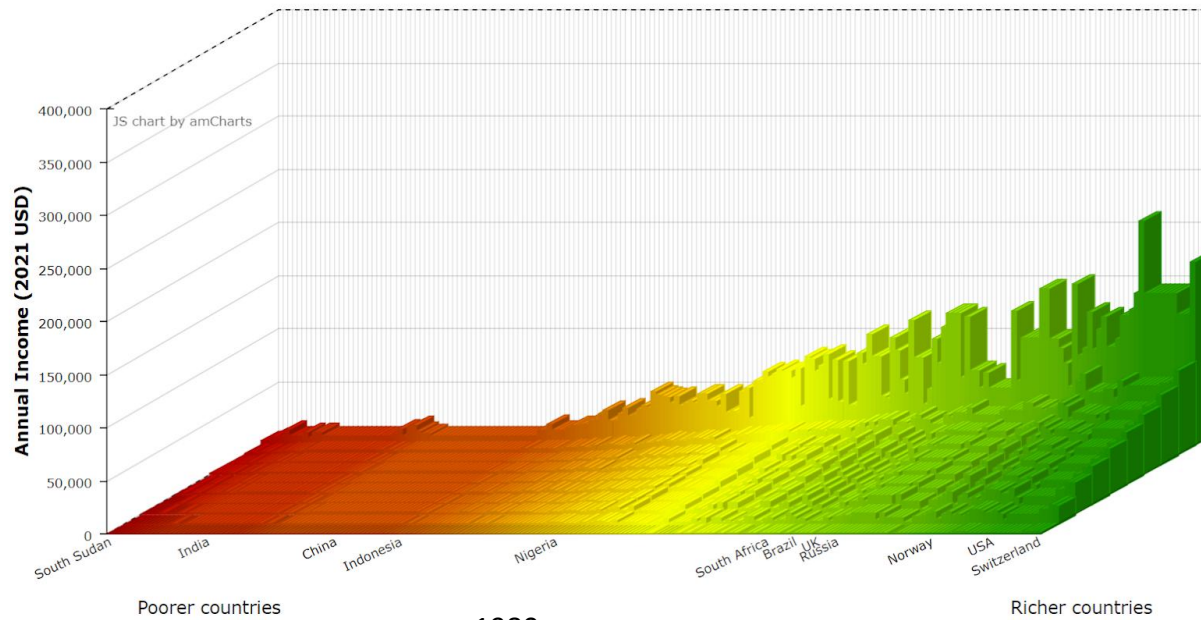
Consequences of climate change:

- Melting of polar ice caps
- Rising sea levels
- Changes in climate and rain patterns
- Destruction of food-growing areas
- Uninhabitable zones

Income inequality

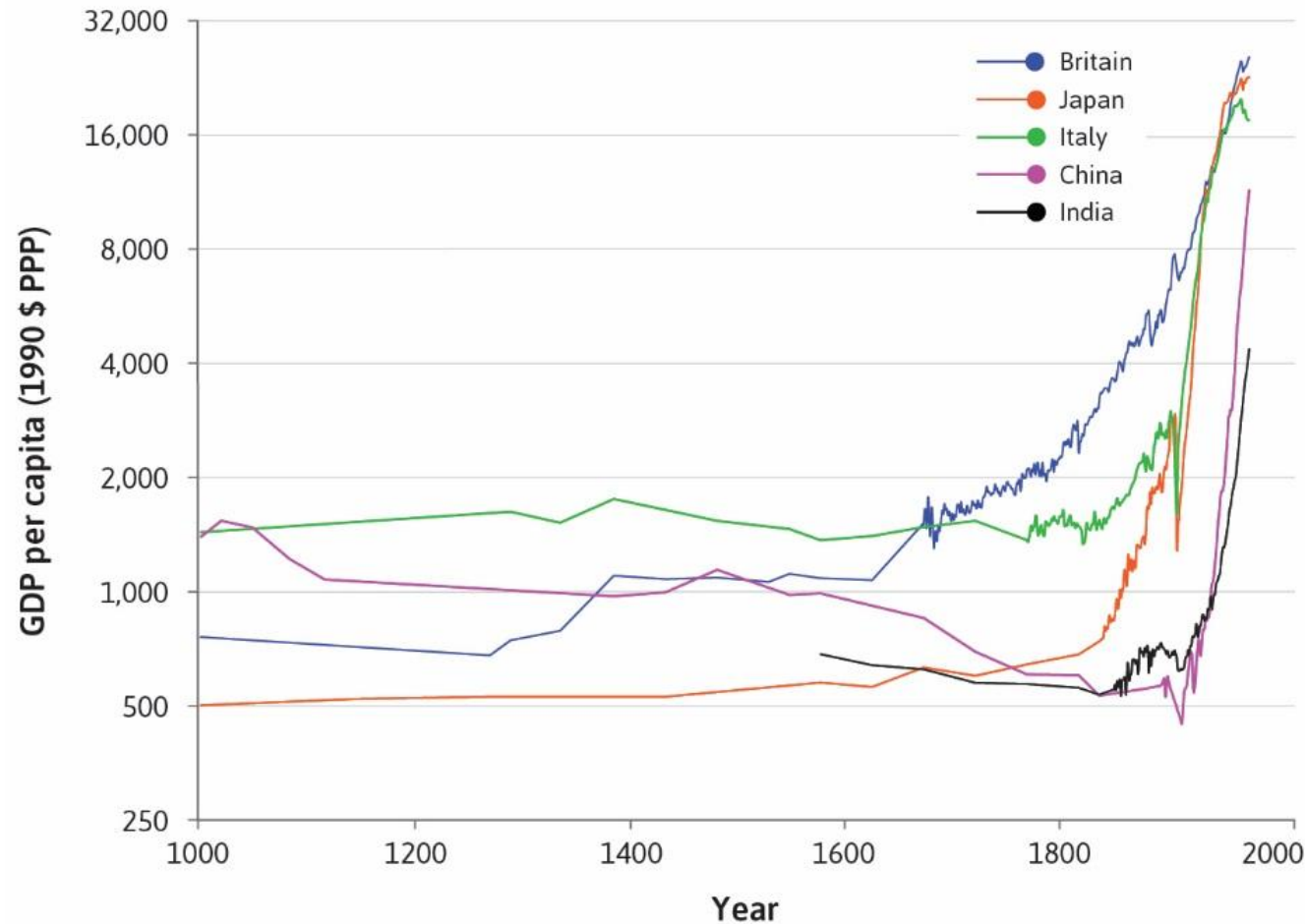


Income inequality



C. Topic 2: Explaining the hockey stick

GDP growth rates



“Hockey-stick” curves represent the sustained rapid growth in GDP per capita experienced by countries worldwide.

Quick question

- Use your Gen AI / LLM assistant to:

1. *Prompt:*

"As a farmer before modern machinery, describe how you plant and harvest crops."

2. *Prompt:*

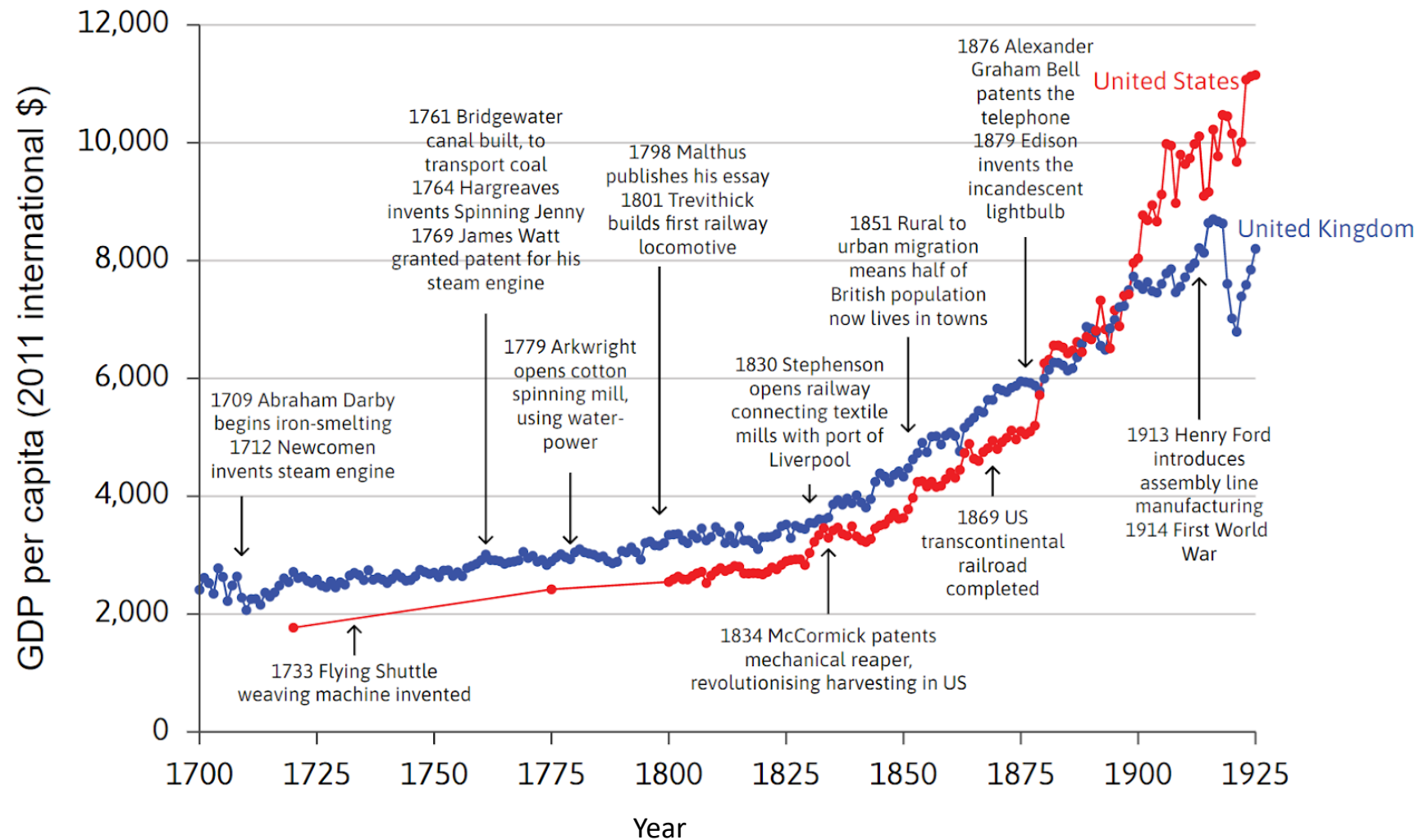
"As a modern farmer using advanced machinery, describe how you plant and harvest crops."

-> Compare the AI-generated narratives and discuss how technology increases productivity and contributes to economic growth (recall the hockey stick pattern).

Continuous technological revolution

- Huge number of scientific and technological advances coincided with the upward kink in the hockey stick in Britain in the middle of the 18th century.
- New technologies emerged in textiles, energy and transportation which themselves became obsolete quite quickly.
- This period of cumulative innovation is called the **Industrial Revolution**.
- **Technology** is a process that takes a set of inputs and creates an output.

Continuous technological revolution

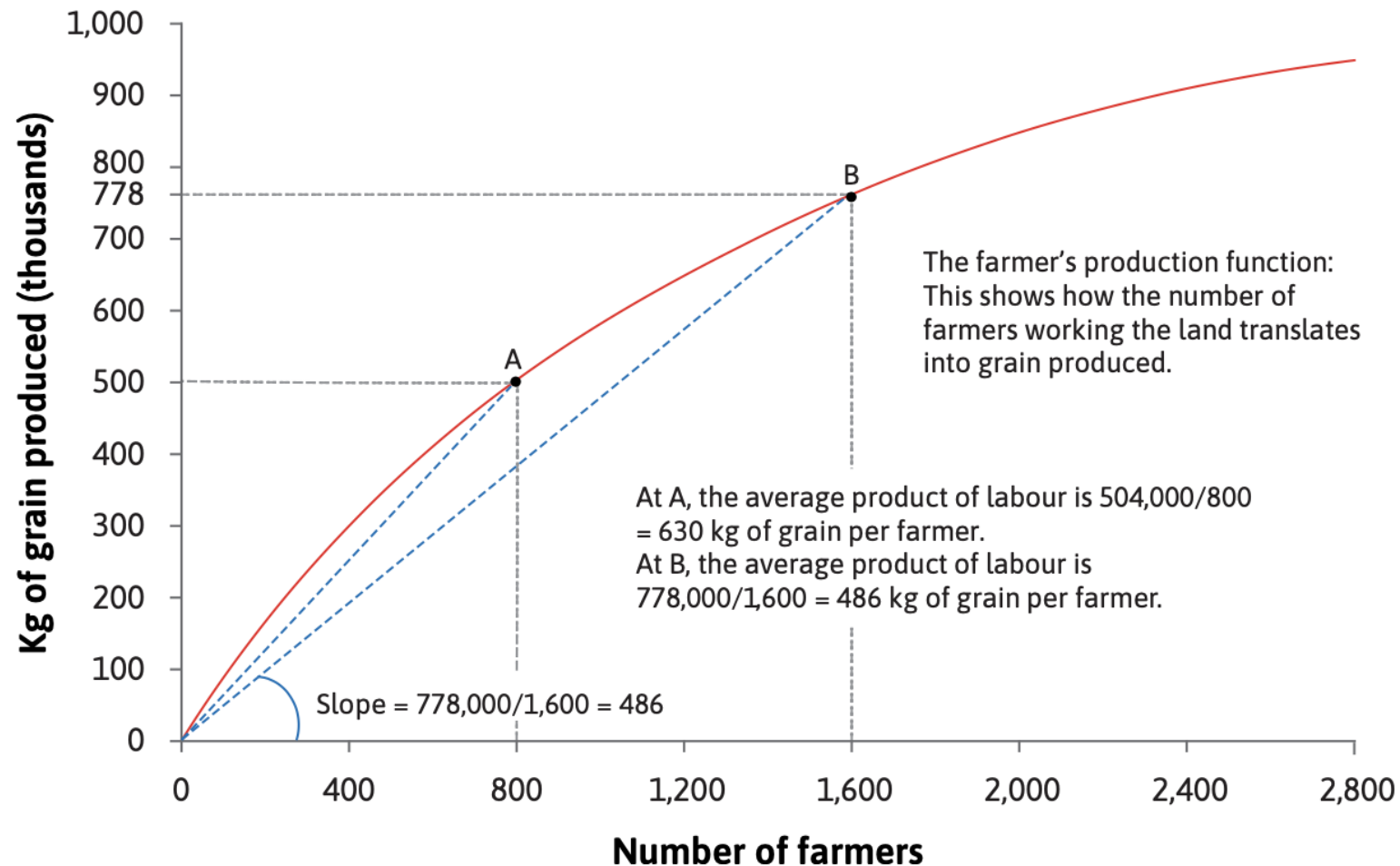


Malthus' model: Explaining the flat part

- A simple model in which food is produced using only labour and land.
- Labour and land are the **factors of production** in the model: the inputs into the production process.
- The relationship between the inputs to a production process and the amount of output is given by a **production function**.
- Malthus believed that in agriculture there was a **diminishing average product of labour** meaning if there is a fixed quantity of land, then average output per worker falls as more workers are added.

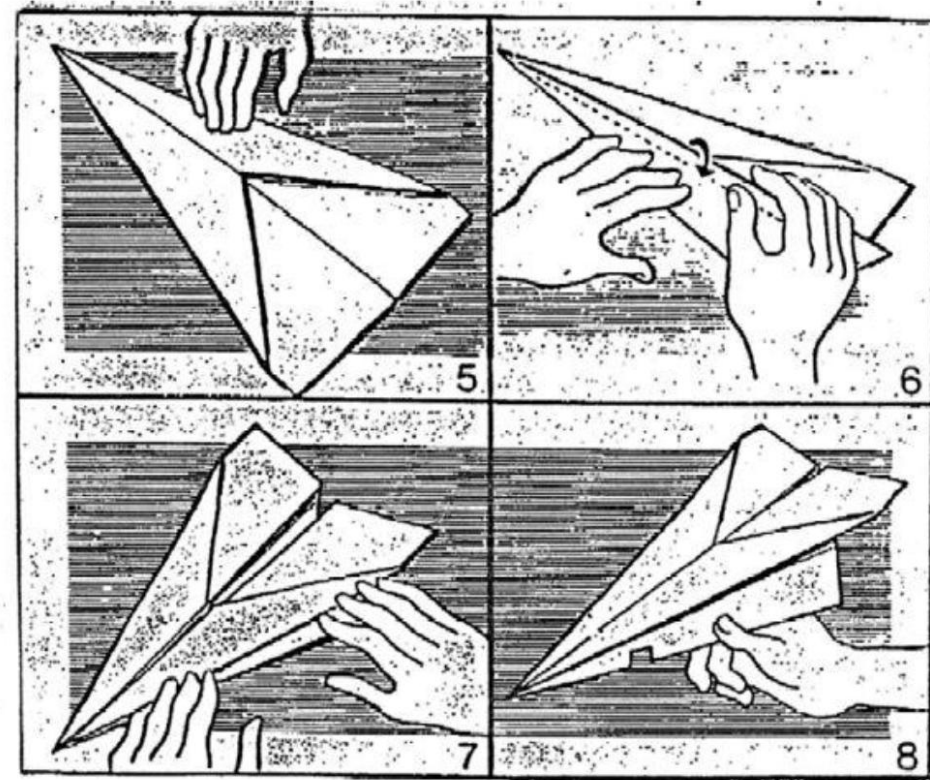
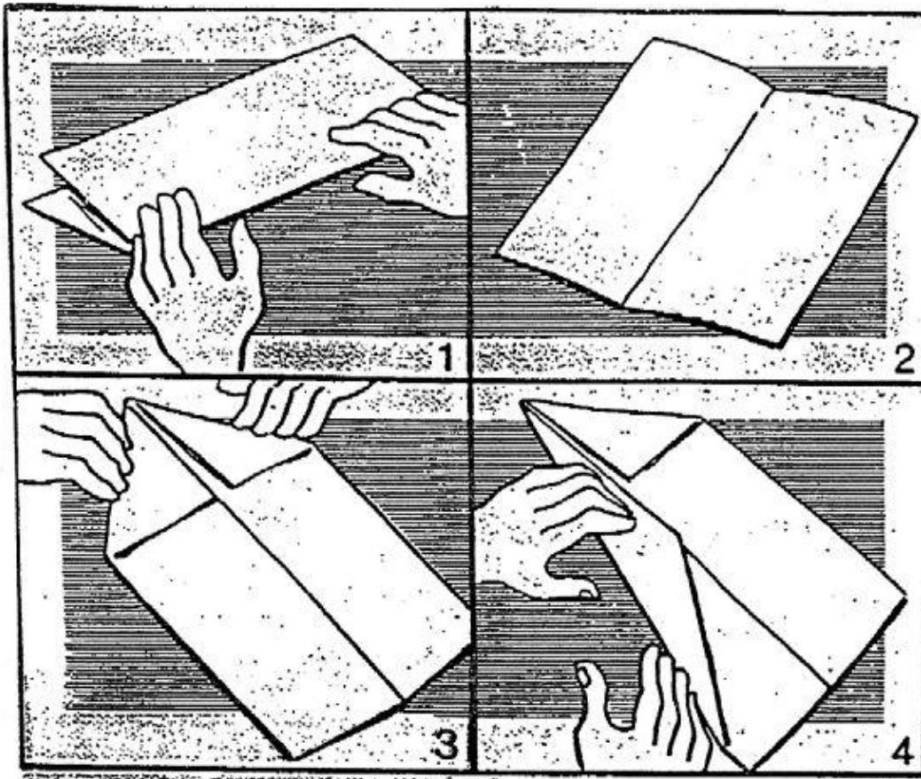
$$\text{average product of labour} = \frac{\text{total output}}{\text{total number of farmers}}$$

The production function



Malthus' model: practical explanation

Images for the approved paper aeroplanes are provided here:



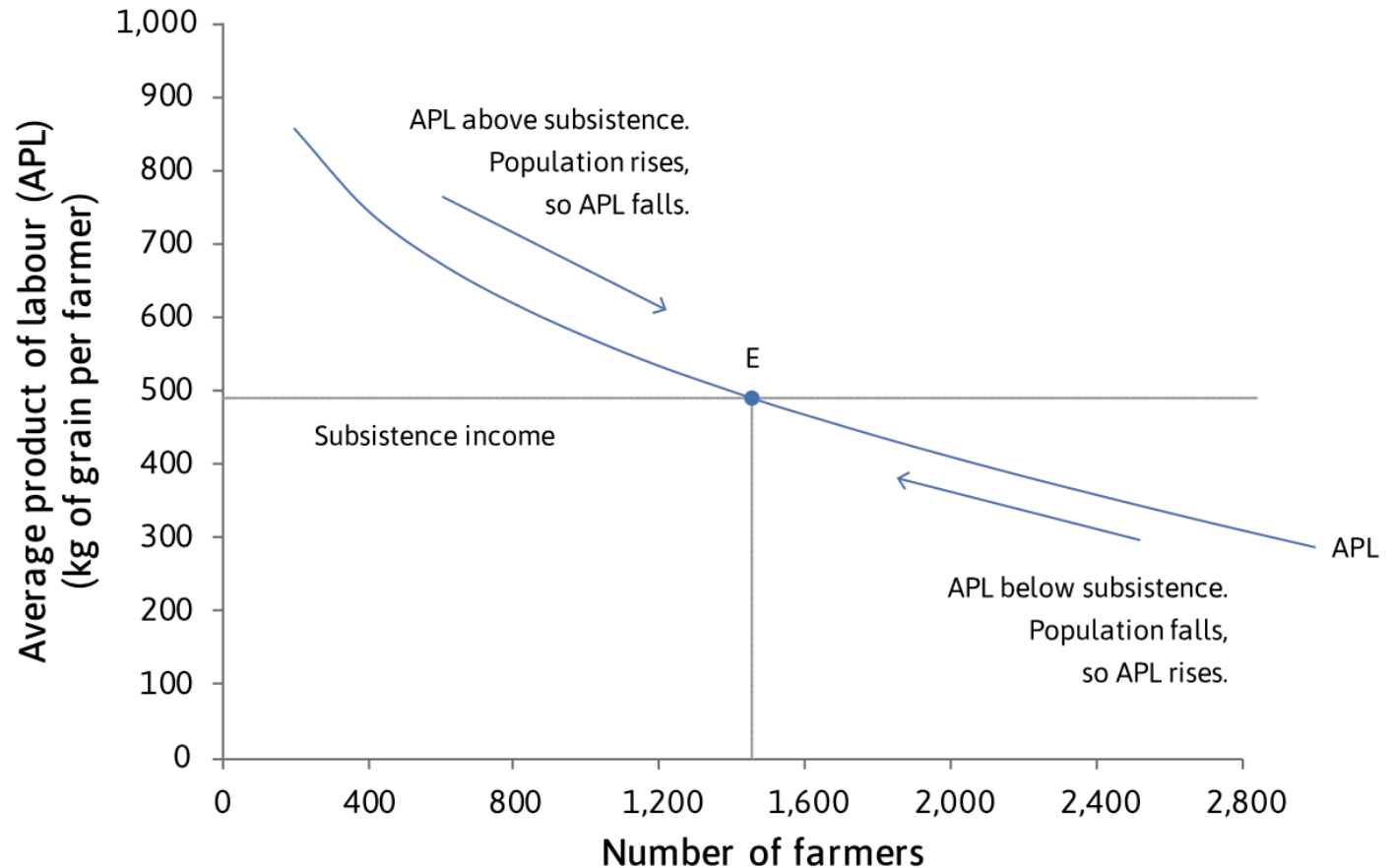
From coreecon

Population and living standards

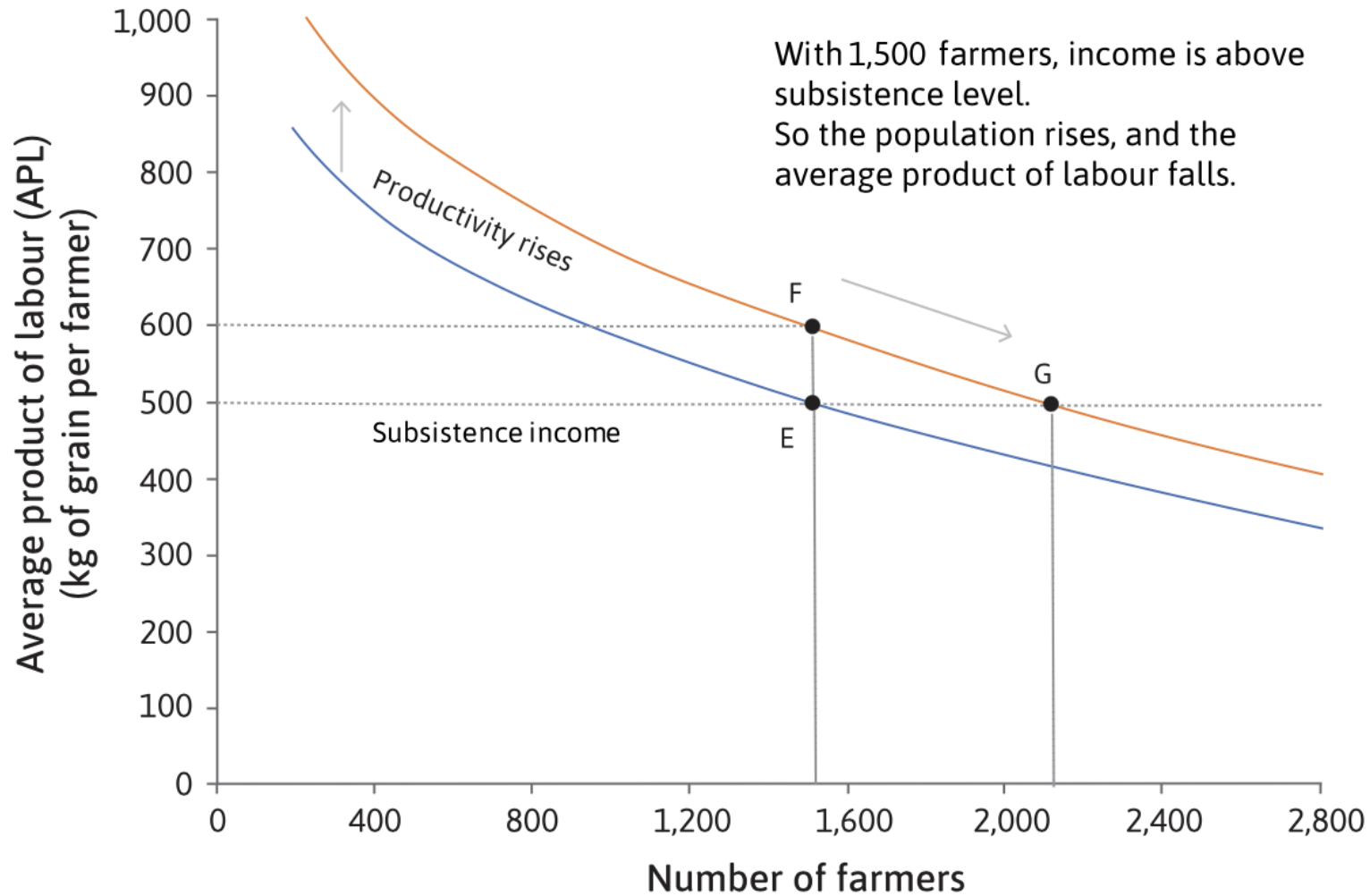
- When people were well-fed they would multiply.
- Population expands if living standards increase.
- Further population growth would push down incomes as a result of diminishing average product of labour.
- Falling living standards would slow population growth.
- Ultimately incomes would settle at a **subsistence level**. This is a level of income where the population remains constant over time.

Equilibrium in the Malthusian model

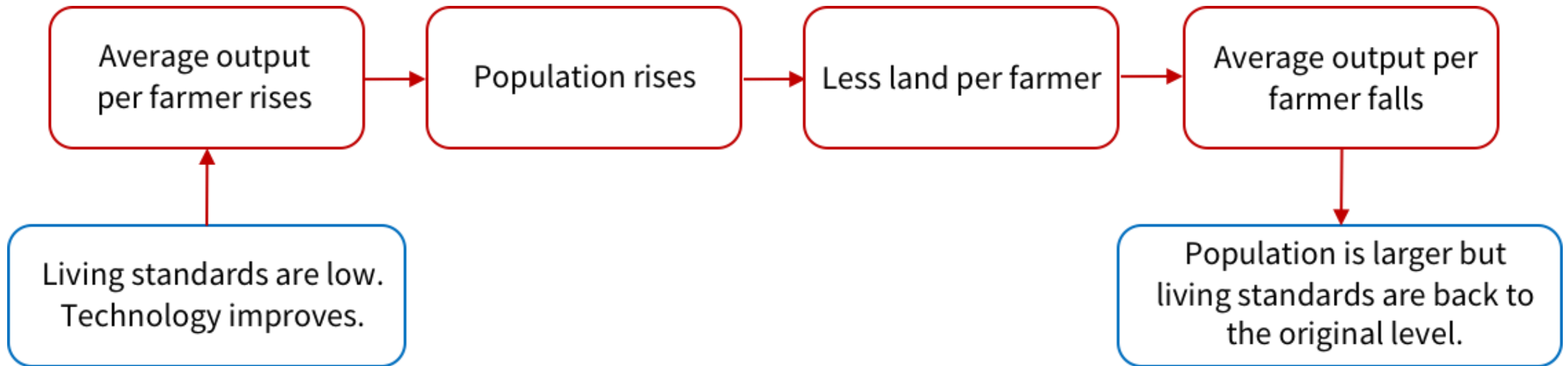
- Equilibrium at 500 kg and a population of 1500 farmers.
- An **equilibrium** is a point where there is no tendency to change unless something else in the economy changes.



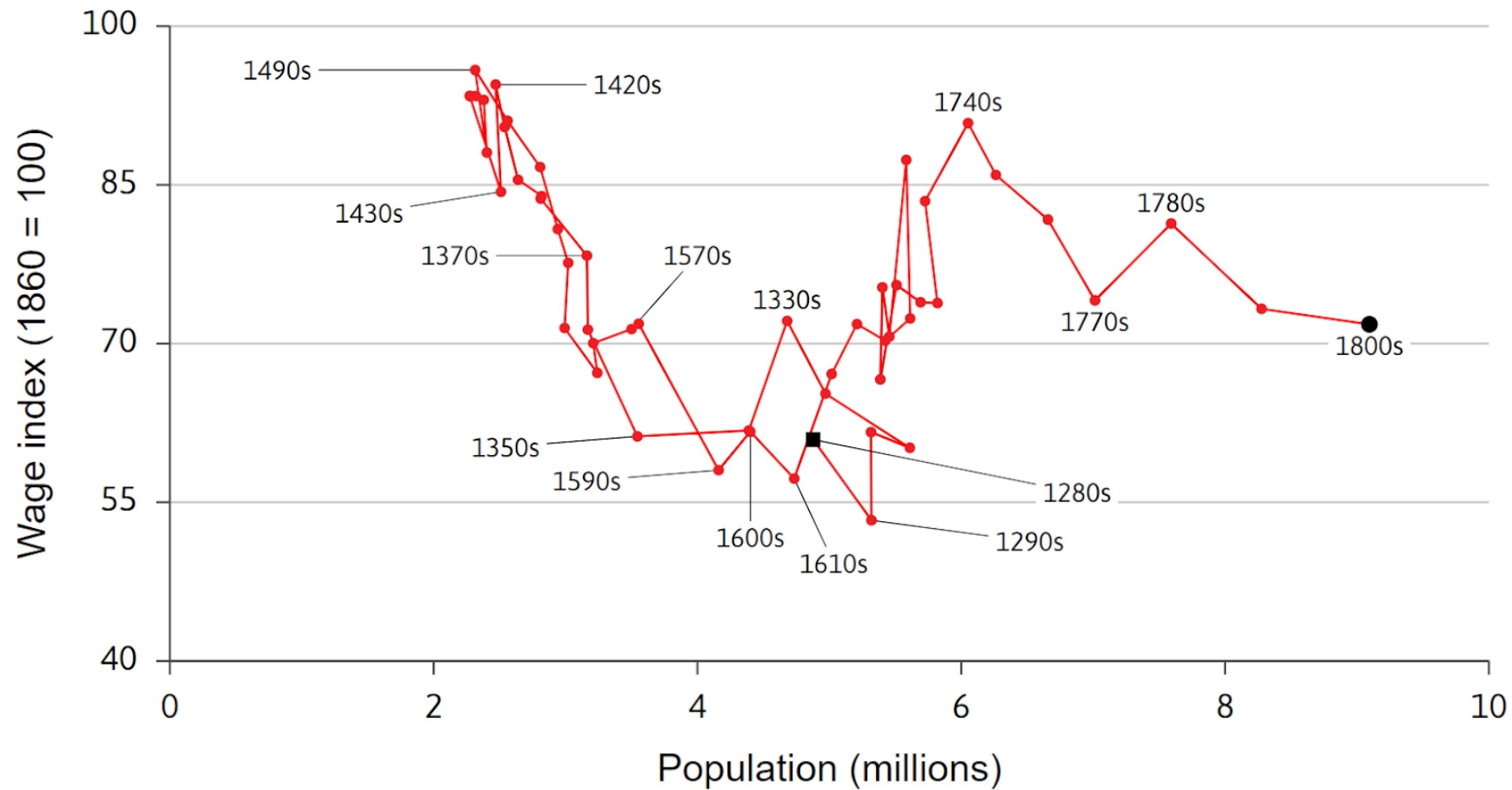
Malthus' Law



A one-off technological improvement



Evidence of Malthus' Law

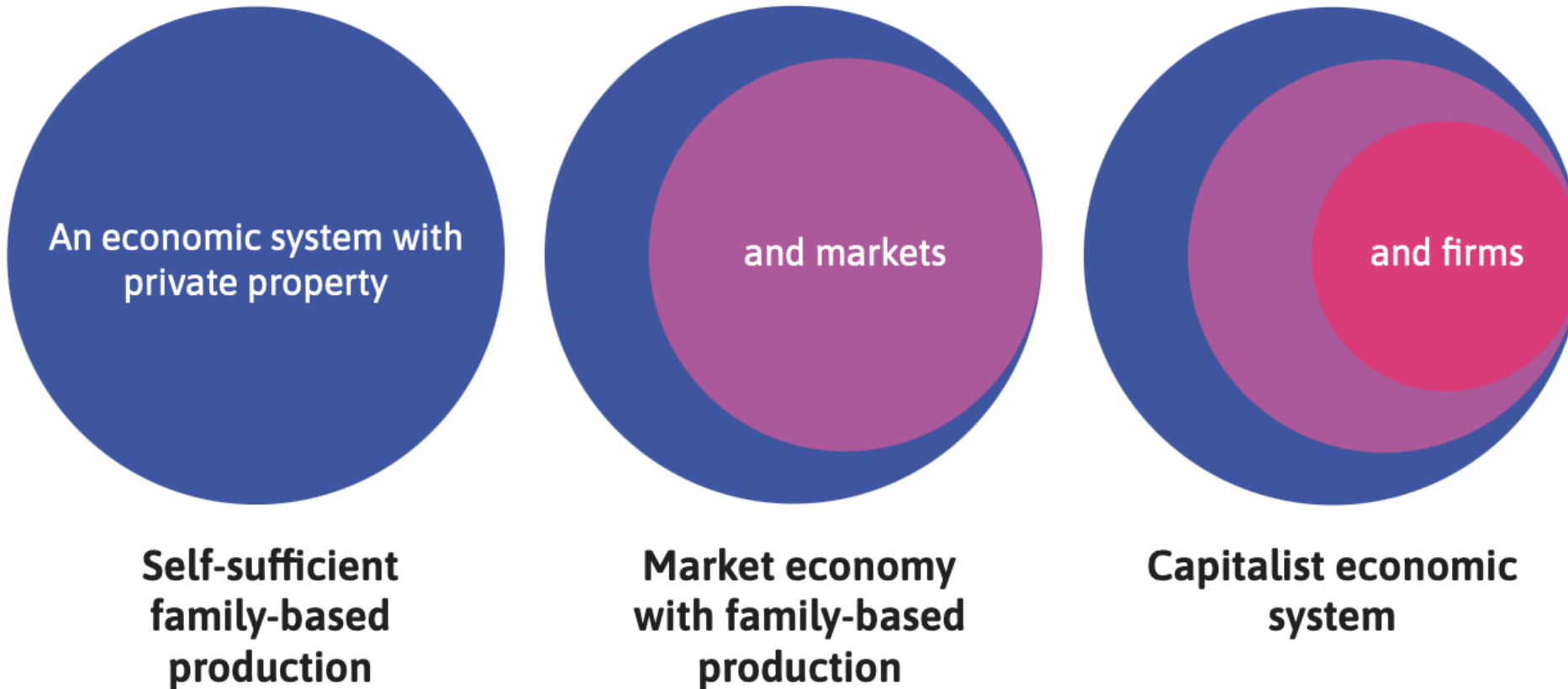


D. Topic 3: Further important considerations

Economic systems and institutions

- **Capitalism** is an **economic system** characterised by a particular combination of **institutions**.
- **Institutions:** the laws and informal rules that regulate social interactions among people and between people and the biosphere.
- **Economic system:** a way of organizing the economy that is distinctive in its basic institutions.

Capitalist institutions



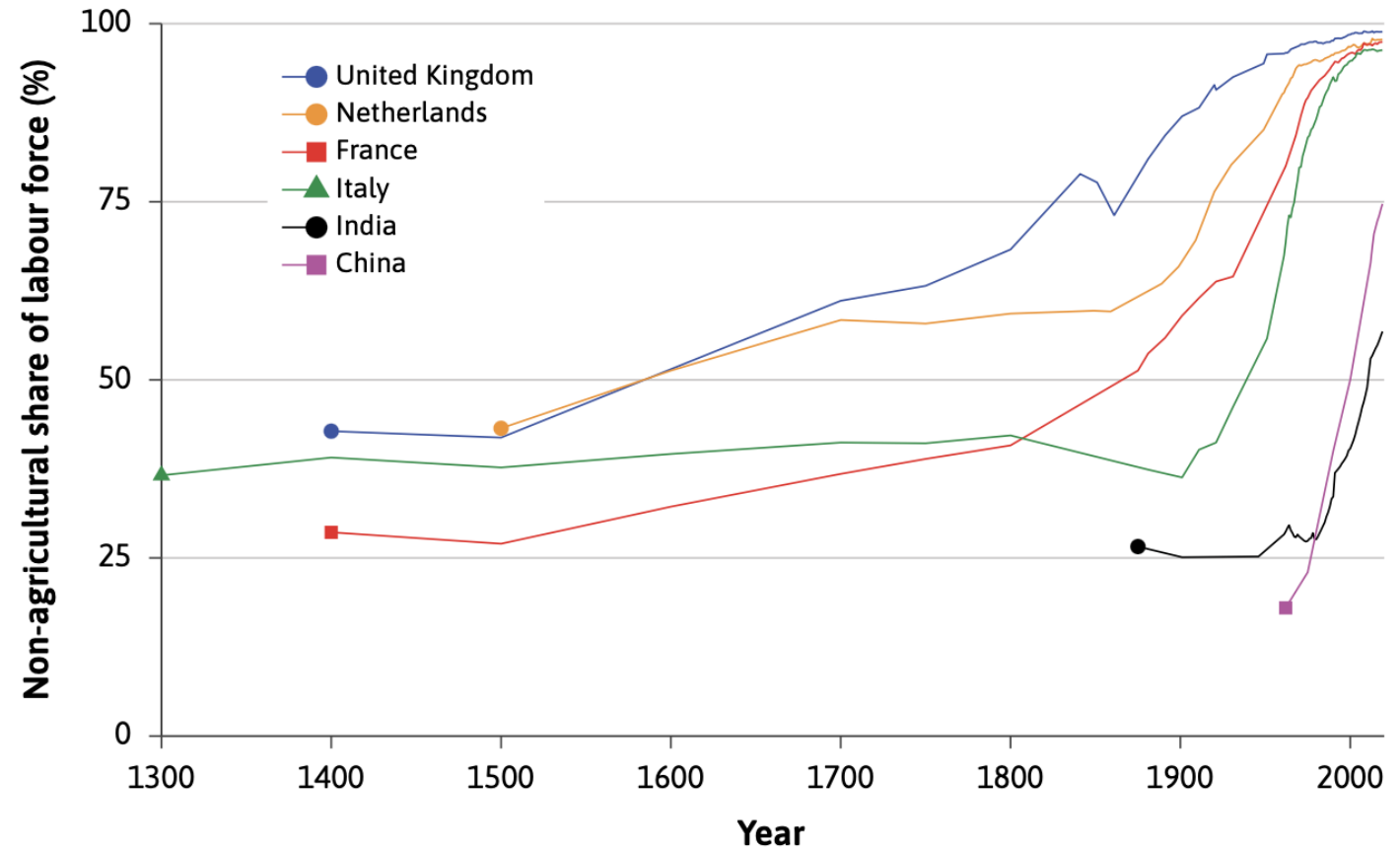
Capitalist institutions

- Read the article, pay a special attention to the term '*capitalism*' and discuss how this term is being used (according to CORE's definition (private property, markets, firms)).
- Notice the author perspective on capitalism, is it positive, negative, mixed or neutral?
- BBC. [Trump's brand of US capitalism faces 'socialist' backlash from conservatives.](#)

Capitalism and living standards

The emergence of capitalism brought two major changes:

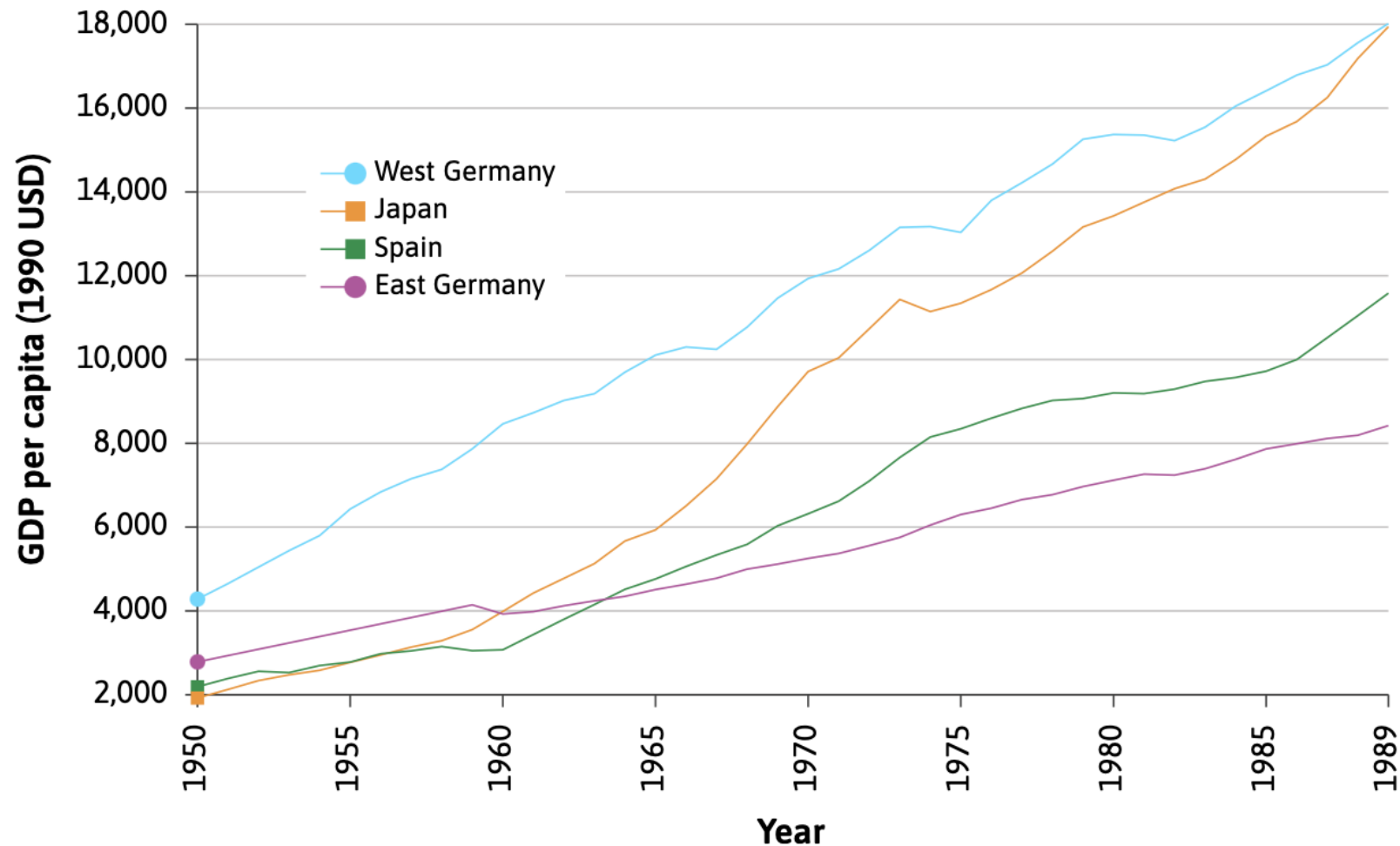
- Technology
- Specialisation



Did capitalism cause the upward kink?

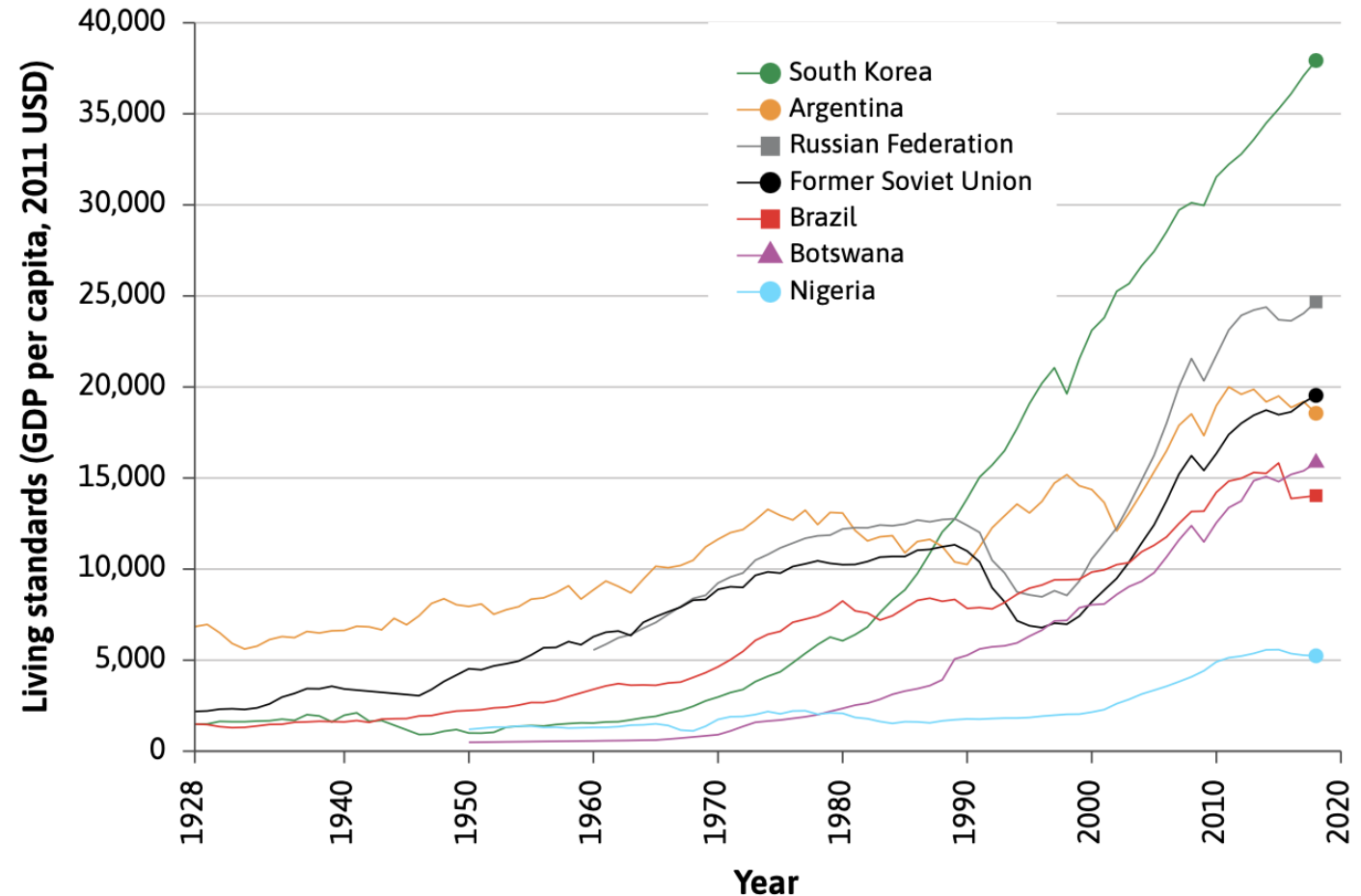
- A **causal statement** is one where we say that change X causes Y to happen.
- Scientists use experiments to test causal statements. This is difficult for economists. We make use of **natural experiments**.
- A **natural experiment** is a situation where we use an external event, such as a change in institutions or a natural disaster, to compare the outcomes for economic actors who were affected by this event and those who were not.
- A **counterfactual** event is what would have happened if there had been no change.

Did capitalism cause the upward kink?



Varieties of capitalism

A **developmental state** is where the government plays a leading role in economic take-off.



Varieties of capitalism

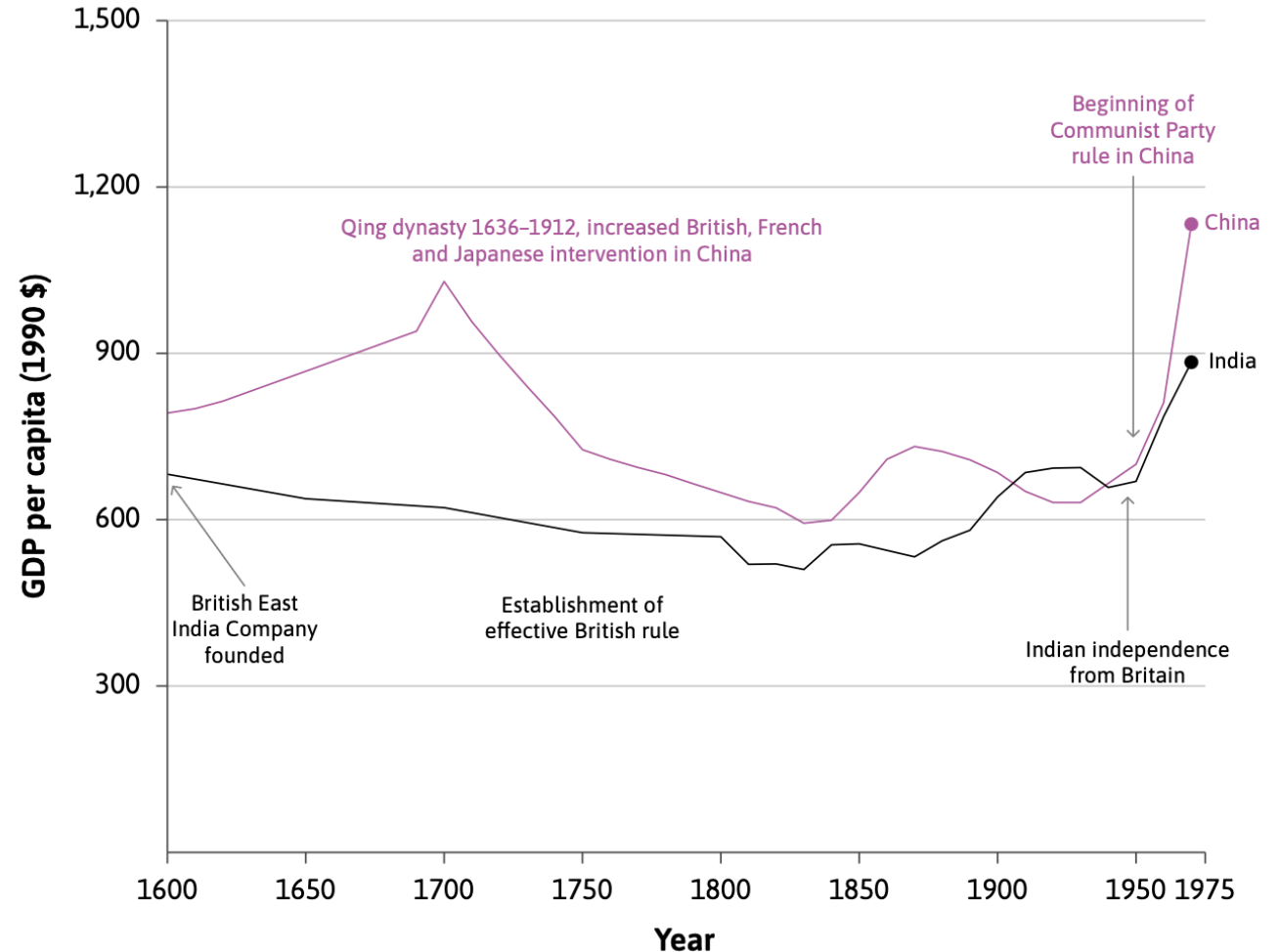
Capitalism is dynamic when:

- Private property is secure
- Markets for firms' output are competitive
- Firms are not controlled by people because of government connections or privileged by birth
- Markets, private property and firms are all regulated by laws and policies
- There is a well functioning legal system and government provides essential goods and services

Colonization and economic growth

In India, colonization reduced living standards because:

- Colonial rulers developed infrastructure for exporting goods, not for the benefit of local producers.
- Colonial rulers established institutions that involved local elites focusing on collecting taxes rather than improving the economy.

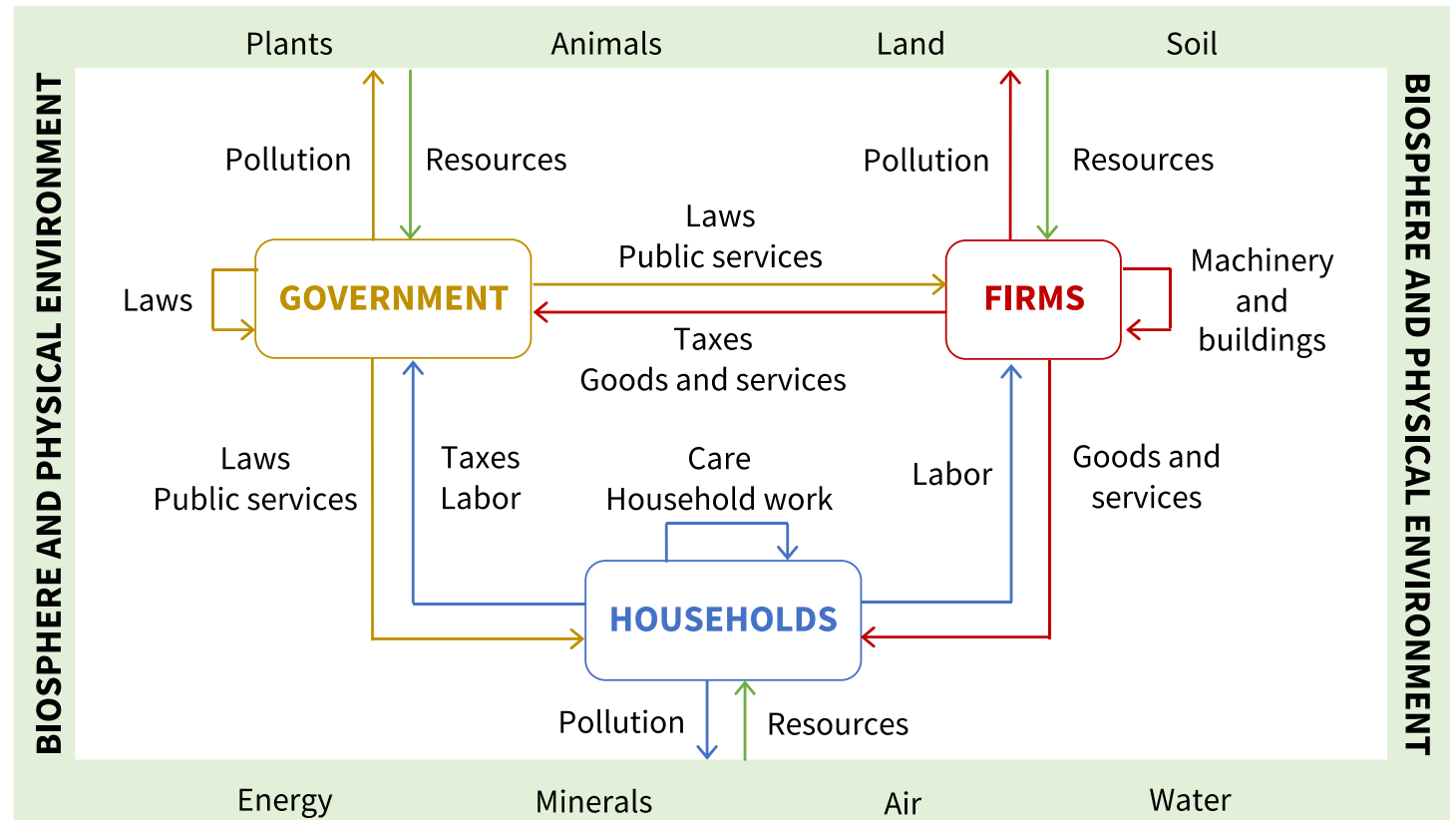
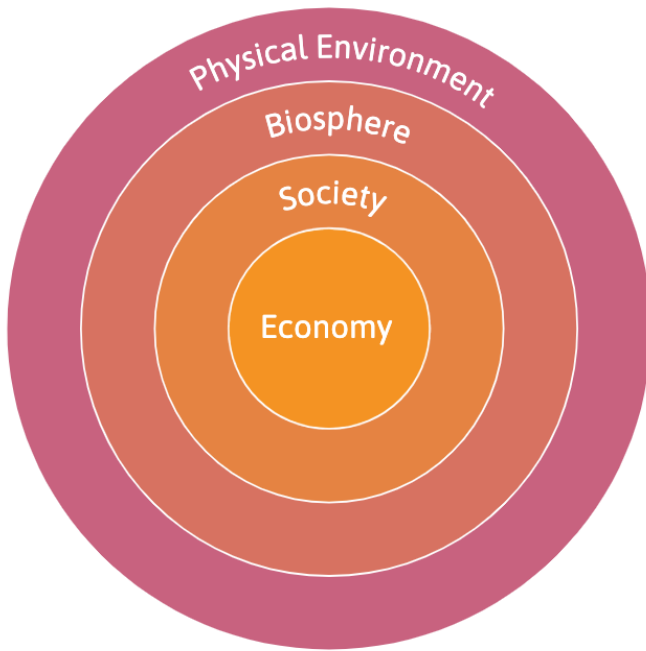


E. Economics and the biosphere

The economy and biosphere

- **Economics** is the study of how people interact with each other and with their natural environment in producing and acquiring their livelihoods, and how this changes over time and differs across societies.
- The **biosphere** is the collection of all forms of life on earth. The physical environment and the biosphere provide the essentials for life such as air, water and food as well as the raw materials that we use in production such as wood, metals and oil.

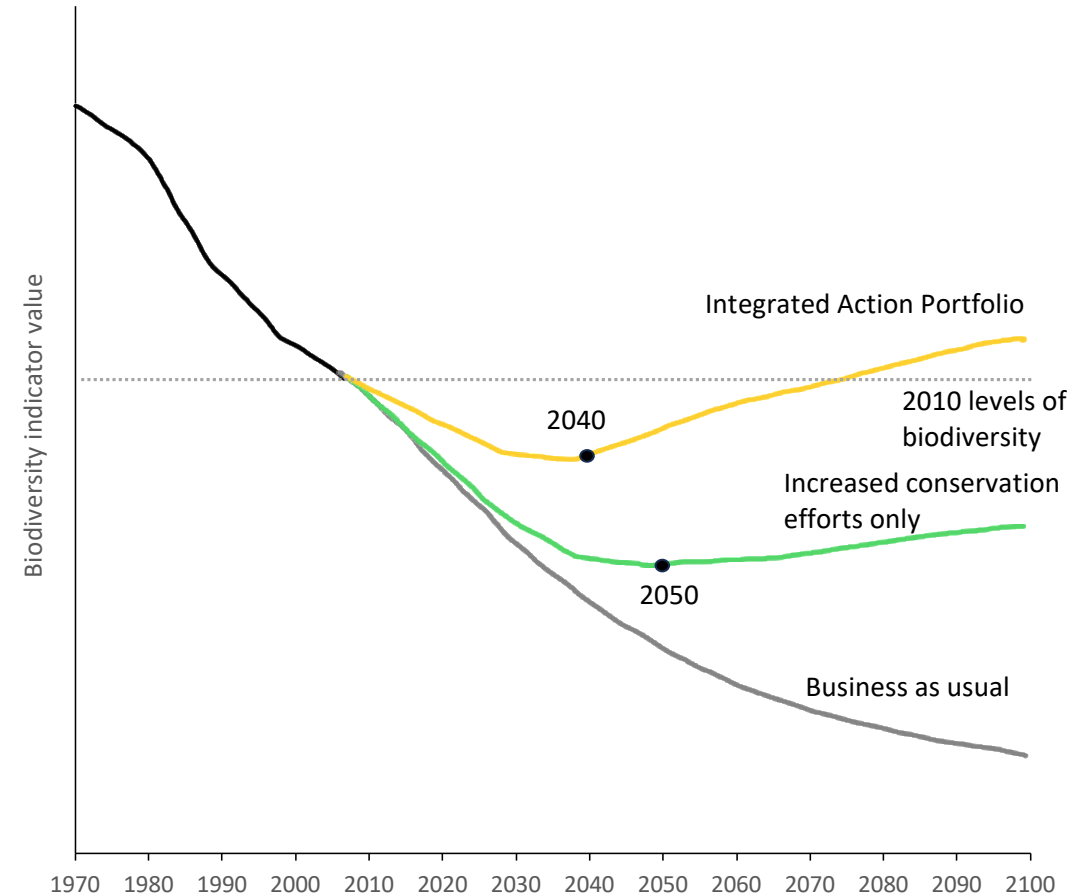
The economy



Economics and environmental issues

Governments can address environmental problems by:

- Regulating the amount of emissions or other environmental damages
- This regulation can be done by banning or issuing a limited number of permits



Summary

1. Important trends in economic variables over time
 - Income inequality across regions has increased a lot over time
 - “Hockey-stick” growth in GDP, and its negative consequences
 - Malthus’ law explains the flat part.
 - **Technological progress** helped bring about the vertical part.
2. The adoption of capitalism was another key factor.
 - **Capitalism = Private property + Markets + Firms**
 - Failure of these institutions and different political systems can explain divergence in economic growth across countries.

In the next unit

- Using economic models to explain the trends in technological growth over time
- The role of firms in technological development